4. Control

4.1 Generalities

Caution

This section applies to firmware 3.0 (grippers delivered after November 2011). For prior versions please see the documentation archives.

The Robotiq Adaptive Gripper S-Model is controlled from the robot controller (see Figure 4.1.1) using an industrial protocol (EtherNet/IP, DeviceNet, CANopen, EtherCat, etc.). The programming of the Gripper can be done with the *Teach Pendant* of the robot or by offline programming.

Info

- For each Operation Mode, the operator can control the force and the speed of the fingers.
- Unless individual control is selected, the fingers movement is always synchronized, movement is done with a single "Go to requested position" command (the motion of each mechanical phalanx is done automatically).

Since the Robotiq Adaptive Gripper S-Model has its own internal controller, high-level commands such as "Go to requested position" are used to control it. The embedded Robotiq Controller takes care of the regulation of the speed and the force prescribed, while the mechanical design of the fingers automatically adapts to the shape of object(s).

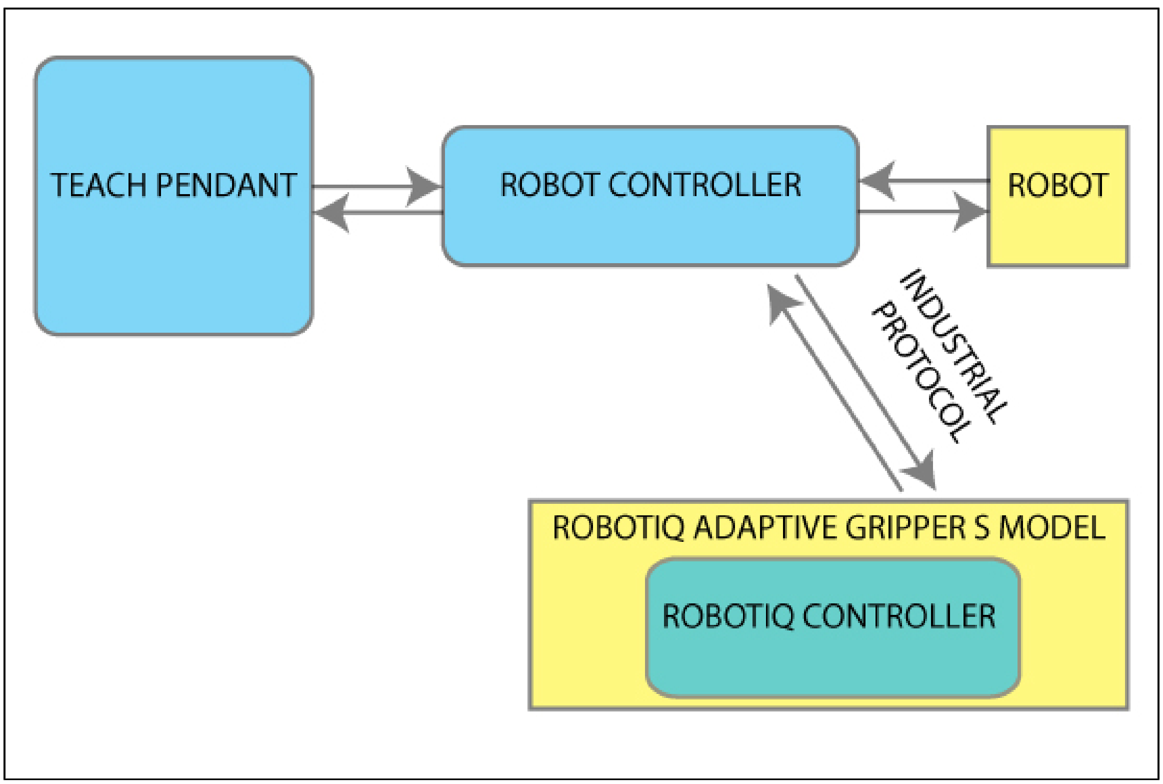

Figure 4.1.1 : Adaptive Gripper S-Model connections.

4.2 Status overview

The Adaptive Gripper S-Model returns several registers of information to the robot controller:

- **Global Gripper Status** - A global Gripper Status is available. This gives information such as which Operation Mode is currently active or if the Gripper is closed or open.
- **Object Status** - There is also an Object Status that let you know if there is an object in the Gripper and, in the affirmative, how many fingers are in contact with it.
- **Fault Status** - The Fault Status gives additional details about the cause of a fault.
- **Position Request Echo** - The Gripper returns the position requested by the robot to make sure that the new command has been received correctly.
- **Motor Encoder Status** - The information of the encoders of the four motors is also available.
- **Current Status** - The current of the motors can also be known. Since the torque of the motor is a linear function of the current, this gives information about the force that is applied at the actuation linkage of the finger.

4.3 Control overview

The Gripper controller has an internal memory that is shared with the robot controller. One part of the memory is for the robot output, **gripper functionalities**. The other part of the memory is for the robot input, **gripper status** (see Figure 4.3.1). Two types of actions can then be done by the robot controller:

1. Write in the **robot output** registers to activate **functionalities**;
2. Read in the **robot input** registers to get the **status** of the gripper.

> **Info**
>
> The Gripper must be initialized (activation bit) at power on. This procedure takes a few seconds and allows the gripper to be calibrated against internal mechanical stops.

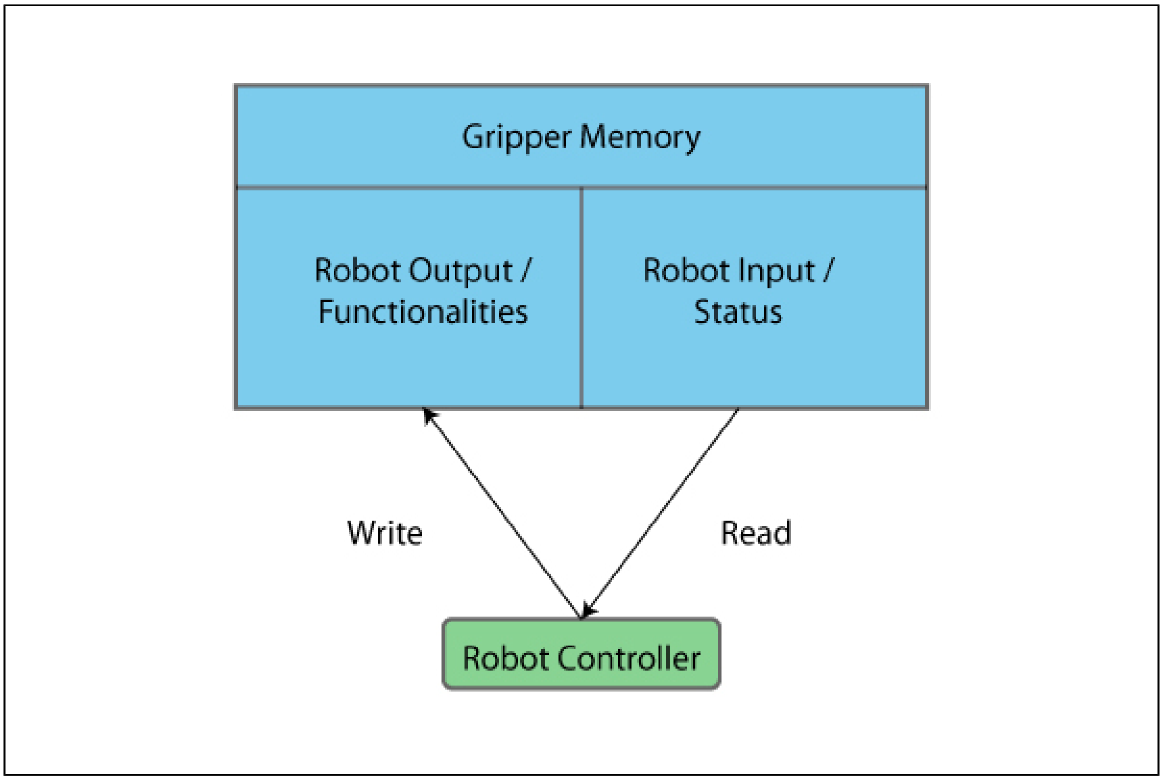

Figure 4.3.1 : Gripper memory shared with the robot controller.

4.4 Status LEDs

Three status LED lights provide general information about the Adaptive Gripper S model status. Figure 4.4.1 shows the LEDs and their locations.

Figure 4.4.1 : Status LEDs.

4.4.1 Supply LED

COLOR	STATE	INFORMATION
Blue	Off	Gripper is not power supplied
Blue	On	The Gripper is correctly supplied and the control board is running

4.4.2 Communication LED

COLOR	STATE	INFORMATION
Green	Off	No network detected
Green	Blinking	A network has been detected and no connection has been established
Green	On	A network has been detected and at least one connection is in the established state

4.4.3 Fault LED

COLOR	STATE	INFORMATION
Red	Off	No fault detected
Red	On	A minor fault occurred (or the Gripper si booting)
Red	Blinking	A major fault occurred

Info

A major fault refers to a situation where the Gripper must be reactivated.

4.5 Gripper register mapping

Caution

This section applies to firmware 3.0 (Grippers delivered after November 2011). For prior versions please see the documentation archives.

Info

Register format is Little Endian (Intel format), namely from LSB (Less Significant Bit) to MSB (Most Significant Bit).

Version 3 of the Adaptive Gripper S-Model firmware provides new functionalities such as the direct position control of the fingers via "go to" commands. There is also additional advanced options such as the individual control of the fingers and scissor and the automatic centering of the fingers.

A Simplified Control Mode is available for users which do not intend to use the advanced option otherwise a register mapping for the Advanced Control Mode containing all the gripper functionalities is also provided. From the gripper standpoint, there is no difference between the two modes. The Simple Control Mode is only intended to ease the usage of the gripper for users who are only interested in the basic functionalities.

Warning

When using the Simplified Control Mode, it is important to fill the unused registers with zeros. Neglecting to do so would result in the unwanted triggering of control options and could lead to a hazardous behavior of the Gripper.

Register mapping for the Simplified Control Mode:

> **Caution**
>
> Byte **numeration starts on zero** and not at 1 for the functionalities and status registers.

REGISTER	ROBOT OUTPUT / FUNCTIONALITIES	ROBOT INPUT / STATUS
Byte 0	ACTION REQUEST	GRIPPER STATUS
Byte 1	00000000	OBJECT DETECTION
Byte 2	00000000	FAULT STATUS
Byte 3	POSITION REQUEST	POS. REQUEST ECHO
Byte 4	SPEED	FINGER A POSITION
Byte 5	FORCE	FINGER A CURRENT
Byte 6	00000000	NOT USED IN SIMPLE MODE
Byte 7	00000000	FINGER B POSITION
Byte 8	00000000	FINGER B CURRENT
Byte 9	00000000	NOT USED IN SIMPLE MODE
Byte 10	00000000	FINGER C POSITION
Byte 11	00000000	FINGER C CURRENT
Byte 12	00000000	NOT USED IN SIMPLE MODE
Byte 13	00000000	SCISSOR POSITION
Byte 14	00000000	SCISSOR CURRENT
Byte 15	RESERVED	RESERVED

Register mapping for the Advanced Control Mode

REGISTER	ROBOT OUTPUT / FUNCTIONALITIES	ROBOT INPUT / STATUS
Byte 0	ACTION REQUEST	GRIPPER STATUS
Byte 1	GRIPPER OPTIONS	OBJECT DETECTION
Byte 2	GRIPPER OPTIONS #2 (EMPTY)	FAULT STATUS
Byte 3	POSITION REQUEST (FINGER A IN INDIVIDUAL MODE)	POS. REQUEST ECHO
Byte 4	SPEED (FINGER A IN INDIVIDUAL MODE)	FINGER A POSITION
Byte 5	FORCE (FINGER A IN INDIVIDUAL MODE)	FINGER A CURRENT
Byte 6	FINGER B POSITION REQUEST	FINGER B POS. REQUEST ECHO
Byte 7	FINGER B SPEED	FINGER B POSITION
Byte 8	FINGER B FORCE	FINGER B CURRENT
Byte 9	FINGER C POSITION REQUEST	FINGER C POS. REQUEST ECHO
Byte 10	FINGER C SPEED	FINGER C POSITION
Byte 11	FINGER C FORCE	FINGER C CURRENT
Byte 12	SCISSOR POSITION REQUEST	SCISSOR POS. REQUEST ECHO
Byte 13	SCISSOR SPEED	SCISSOR POSITION
Byte 14	SCISSOR FORCE	SCISSOR CURRENT
Byte 15	RESERVED	RESERVED

4.6 Robot output registers & functionalities

Caution

This section applies to firmware 3.0 (Grippers delivered after November 2011). For prior versions please see the documentation archives.

Info

Register format is Little Endian (Intel format), namely from LSB (Less Significant Bit) to MSB (Most Significant Bit).

Register: ACTION REQUEST
Address: Byte 0

BIT	NAME	DESCRIPTION
0	rACT	0 – Reset Gripper 1 – Activate Gripper (Must stay on after activation routine is completed)
1	rMOD	00 – Go to Basic Mode 10 – Go to Pinch Mode 01 – Go to Wide Mode 11 – Go to Scissor Mode
2		
3	rGTO	0 – Stop 1 – Go to Requested Position
4	rATR	0 – Normal 1 – Automatic release
5-7	rRS0	Reserved

rACT: First action to be made prior to any other actions, **rACT** bit will initialize the Adaptive Gripper. Clear **rACT** to reset Gripper and fault status.

Caution

rACT bit must stay on afterwards for any other action to be performed.

rMOD: Changes the Gripper Grasping Mode. When the Grasping Mode is changed, the Gripper first opens completely to avoid interferences between the fingers then go to the selected mode. This option is ignored if the bit **rICS** is set (individual control of the scissor motion option).

rGTO: The "Go To" action moves the Gripper fingers to the requested position using the configuration defined by the other registers and the **rMOD** bits. The only motions performed without the **rGTO** bit are the activation, the mode change and the automatic release routines.

rATR: Automatic Release routine action slowly open the Gripper fingers until all motions axes reach their mechanical limits. After the motion is completed, the Gripper sends a fault signal and needs to be reinitialized before any other motion is performed. The **rATR** bit overrides all other commands excluding the activation bit (**rACT**).

> **Caution**
>
> The Automatic Release is meant to disengage the Gripper after an emergency stop of the robot. The Automatic Release is not intended to be used under normal operating conditions.

Register: GRIPPER OPTIONS
Address: Byte 1

rAAC: The Automatic Centering option synchronizes the Gripper fingers in order to automatically center the object it seizes. This option requires that fingers B and C have the same position request and velocity. It is not intended to be used in the scissor mode. **This option is currently in a beta version and may be modified in future versions of the firmware**.

rICF: In Individual Control of Fingers mode each finger receives its own command (position request, speed and force) unless the Gripper is in the Scissor Grasping Mode and the Independent Control of Scissor (**rICS**) is not activated. Please refer to the **rPRA** (Position Request) register description for information about the reachable positions of the fingers.

> **Caution**
>
> As soon as the **rICF** bit is set, the fingers will move towards the target defined by the position request bytes. To avoid unwanted motion of the fingers, it is preferable to define the position requests before setting the **rICF** bit. It is also possible to clear the **rGTO** bit, configure the registers according to the desired motion and then set the **rGTO** bit to start the motion.

BIT	NAME	DESCRIPTION
0	rGLV	Reserved
1	rAAC	0 – Normal 1 – Enable Automatic Auto-Centering
2	rICF	0 – Normal 1 – Enable Individual Control of Fingers A, B and C
3	rICS	0 – Normal 1 – Enable Individual Control of Scissor. Disable Mode Selection.
4-7	rRS1	Reserved

rICS: In Individual Control of Scissor the scissor axis moves independently from the Grasping mode. When this option is selected, the **rMOD** bits (Grasping Mode) are ignored as the scissor axis position is defined by the **rPRS** (Position Request for the Scissor axis) register.

> **Caution**
>
> To avoid geometrical interference between fingers B and C, the reachable positions for the scissor axis is reduced if the Individual Control of Scissor option is selected. Please refer to the **rPRA** (Position Request) register description for more information about the reachable positions of the scissor axis.

Register: GRIPPER OPTIONS 2
Address: Byte 2

BIT	NAME	DESCRIPTION
0 – 7	rRS2	Reserved

Register: POSITION REQUEST (FINGER A IN INDIVIDUAL MODE)
Address: Byte 3

BIT	NAME	DESCRIPTION
0 – 7	rPRA	Set Position Request for the Gripper (finger A in individual mode). 0x00 (Minimum position) to 0xFF (Maximum position)

This register is used to set the Adaptive Gripper fingers target position (or finger A only if bit **rICF** is set). The positions 0x00 and 0xFF correspond respectively to the fully opened and fully closed mechanical stops. Figure 4.6.1 represents the reachable workspace of the fingers and scissor axis. Note that the finger position on the figure represents the maximum value for the three fingers. Also, note that the fully opened and fully closed software limits are not shown on the figure for simplicity. The fully closed software limit of the scissor axis when the Individual Control of Scissor option is selected is also not shown for simplicity.

> **Caution**
>
> In order to protect the Gripper from geometric interferences, several software limits are implemented and therefore some positions are not reachable. When a finger reaches the software limit, the Gripper status will indicate that the requested position was reached. This is because the requested position is internally replaced by the software limit.

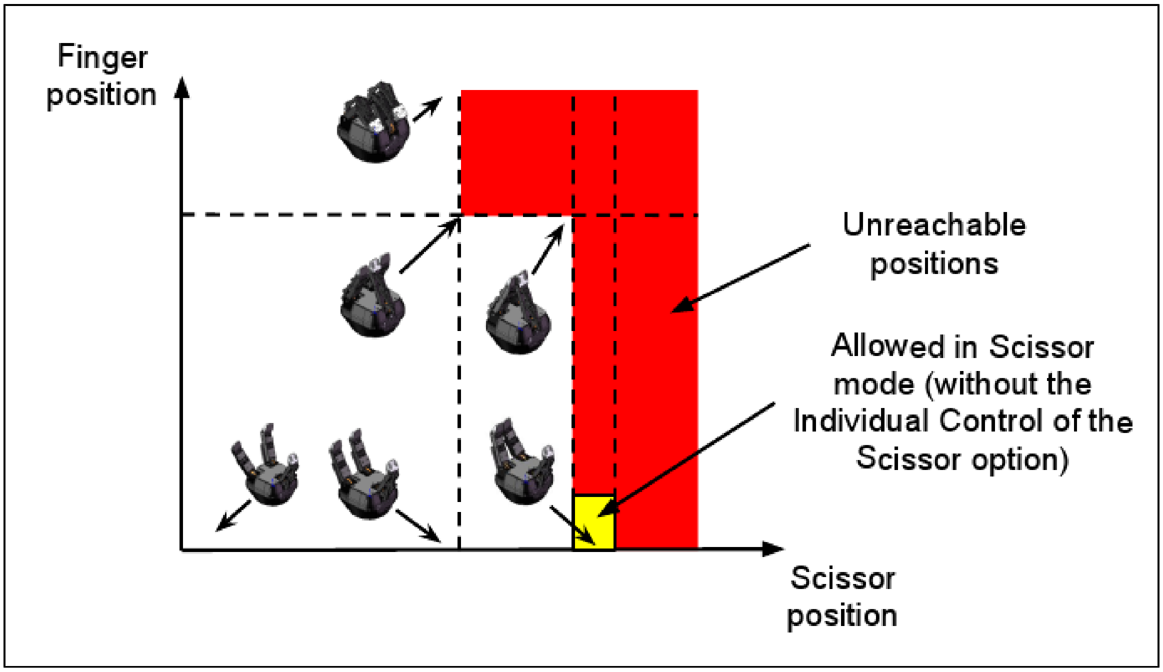

Figure 4.6.1 : Reachable workspace of the fingers and scissor axis.

Register: SPEED (FINGER A IN INDIVIDUAL MODE)
Address: Byte 4

BIT	NAME	DESCRIPTION
0 – 7	rSPA	Set Grasping Speed of the Gripper (finger A in individual mode). 0x00 (Minimum velocity) to 0xFF (Maximum velocity)

This register is used to setup the Gripper closing or opening speed (or finger A only if bit **rICF** is set) in real time, however, setting a speed will not initiate a motion.

Info

0x00 speed does not mean absolute zero speed. It is the minimum speed of the Gripper.
Minimum speed: 22 mm/s
Maximum speed: 110 mm/s
Speed / count: 0.34 mm/s

Register: FORCE (FINGER A IN INDIVIDUAL MODE)
Address: Byte 5

BIT	NAME	DESCRIPTION
0 – 7	rFRA	Set Gripping Force 0x00 (Minimum force) to 0xFF (Maximum force)

The the force setting defines the final grasping force of the Adaptive Gripper (or finger A only if bit **rICF** is set). The force will fix maximum current sent to the motors while in motion. For each finger, if the current limit is exceeded, the finger stops and triggers an object detection notification.

> **Info**
>
> Force setting is overridden for a small distance when the motion is initiated. Also, note that 0x00 force does not mean zero force; it is the minimum force that the Gripper can apply.
> Minimum force: 15 N
> Maximum force: 60 N
> Force / count: 0.175 N (approximate value, relation non-linear)

Register: FINGER B POSITION REQUEST
Address: Byte 6

BIT	NAME	DESCRIPTION
0 – 7	rPRB	Set Position Request for finger B. 0x00 (Minimum position) to 0xFF (Maximum position)

This register is used to set the finger B target position. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rPRA** (position request) register for more information.

Register: FINGER B SPEED
Address: Byte 7

BIT	NAME	DESCRIPTION
0 – 7	rSPB	Set Grasping Speed for finger B. 0x00 (Minimum velocity) to 0xFF (Maximum velocity)

This register is used to set finger B speed. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rSPA** (speed) register for more information.

Register: FINGER B FORCE
Address: Byte 8

BIT	NAME	DESCRIPTION
0 – 7	rFRB	Set Gripping Force for finger B. 0x00 (Minimum force) to 0xFF (Maximum force)

This register is used to set finger B force. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rFRA** (force) register for more information.

Register: FINGER C POSITION REQUEST
Address: Byte 9

BIT	NAME	DESCRIPTION
0 – 7	rPRC	Set Position Request for finger C. 0x00 (Minimum position) to 0xFF (Maximum position)

This register is used to set the finger C target position. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rPRA** (position request) register for more information.

Register: FINGER C SPEED
Address: Byte 10

BIT	NAME	DESCRIPTION
0 – 7	rSPC	Set Grasping Speed for finger C. 0x00 (Minimum velocity) to 0xFF (Maximum velocity)

This register is used to set finger C speed. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rSPA** (speed) register for more information.

Register: FINGER C FORCE
Address: Byte 11

BIT	NAME	DESCRIPTION
0 – 7	rFRC	Set Gripping Force 0x00 (Minimum force) to 0xFF (Maximum force)

This register is used to set finger C force. It is only considered if the Individual Control of Finger option is selected (bit **rICF** is set). Please refer to **rFRA** (force) register for more information.

Register: SCISSOR POSITION REQUEST
Address: Byte 12

BIT	NAME	DESCRIPTION
0 – 7	rPRS	Set Position Request for the scissor axis. 0x00 (Minimum position) to 0xFF (Maximum position)

This register is used to set the scissor axis target position. It is only considered if the Individual Control of Scissor option is selected (bit **rICS** is set). Please refer to **rPRA** (position request) register for more information.

Register: SCISSOR SPEED
Address: Byte 13

BIT	NAME	DESCRIPTION
0 – 7	rSPS	Set Grasping Speed for the scissor axis. 0x00 (Minimum velocity) to 0xFF (Maximum velocity)

This register is used to set the scissor axis speed. It is only considered if the Individual Control of Scissor option is selected (bit **rICS** is set). Please refer to **rSPA** (speed) register for more information.

Register: SCISSOR FORCE
Address: Byte 14

BIT	NAME	DESCRIPTION
0 – 7	rFRS	Set Gripping Force for the scissor axis 0x00 (Minimum force) to 0xFF (Maximum force)

This register is used to set the scissor axis force. It is only considered if the Individual Control of Scissor option is selected (bit **rICS** is set). Please refer to **rFRA** (force) register for more information.

4.7 Robot input registers & status

Caution

This section applies to firmware 3.0 (grippers delivered after November 2011). For prior versions please see the documentation archives.

Info

Register format is Little Endian (Intel format), namely from LSB (Less Significant Bit) to MSB (Most Significant Bit).

Register: GRIPPER STATUS
Address: Byte 0

BIT	NAME	DESCRIPTION
0	gACT	Initialization status (Echo of the rACT bit (Activation bit)): 0 – Gripper reset 1 – Gripper activation
1 2	gMOD	Echo of the rMOD bits (Grasping Mode request) 00 – Basic Mode 10 – Pinch Mode 01 – Wide Mode 11 – Scissor Mode
3	gGTO	Echo of the rGTO bit (Go to bit): 0 – Stopped (or performing activation/grasping mode change/automatic release) 1 – Go to Position Request
4 5	gIMC	00 – Gripper is in reset (or automatic release) state. see Fault Status if Gripper is activated. 10 – Activation in progress. 01 – Mode change in progress. 11 – Activation and mode change are completed.
6 7	gSTA	00 – Gripper is in motion towards requested position (only meaningful if gGTO = 1) 10 – Gripper is stopped. One or two fingers stopped before requested position 01 – Gripper is stopped. All fingers stopped before requested position 11 --Gripper is stopped. All fingers reached requested position

Register: OBJECT STATUS
Address: Byte 1

BIT	NAME	DESCRIPTION
0	gDTA	00 – Finger A is in motion (only meaningful if gGTO = 1) 10 – Finger A has stopped due to a contact while opening 01 – Finger A has stopped due to a contact while closing 11 – Finger A is at requested position
1		
2	gDTB	00 – Finger B is in motion (only meaningful if gGTO = 1) 10 – Finger B has stopped due to a contact while opening 01 – Finger B has stopped due to a contact while closing 11 – Finger B is at requested position
3		
4	gDTC	00 – Finger C is in motion (only meaningful if gGTO = 1) 10 – Finger C has stopped due to a contact while opening 01 – Finger C has stopped due to a contact while closing 11 – Finger C is at requested position
5		
6	gDTS	00 – Scissor is in motion (only meaningful if gGTO = 1) 10 – Scissor has stopped due to a contact while opening 01 – Scissor has stopped due to a contact while closing 11 – Scissor is at requested position.
7		

When a contact is detected, the corresponding axis will stop until one of these conditions is met: a new position request is commanded in the opposite direction, the requested force level is increased or the **rGTO** bit is cleared and set again.

Warning

Resetting the contact detection repeatedly at high frequency using the **rGTO** bit may cause a major failure of the Gripper. This is not considered a normal usage of the Gripper and it is not recommended by Robotiq.

Caution

The object detection is precise only to the order of a few mm. In some circumstances object detection may not detect an object even if it is successfully grasped. For example, picking up a thin object in a fingertip grip may be successful without object detection occurring. For such reasons, use this feature with caution. In such applications the "Gripper is stopped" status of register **gSTA** is sufficient to proceed to the next step of the routine.

Register: FAULT STATUS

Address: Byte 2

BIT	NAME	DESCRIPTION
0 – 3	gFLT	0x00 – No Fault **Priority Fault** 0x05 – Action delayed, activation (reactivation) must be completed prior to action 0x06 – Action delayed, mode change must be completed prior to action 0x07 – The activation bit must be set prior to action **Minor Fault (red LED continuous)** 0x09 – The communication chip is not ready (may be booting) 0x0A – Changing mode fault, interferences detected on Scissor (for less than 20 sec) 0x0B – Automatic release in progress **Major Fault (red LED blinking) – Reset is required** 0x0D – Activation fault, verify that no interference or other error occured 0x0E – Changing mode fault, interferences detected on Scissor (for more than 20 sec) 0x0F – Automatic release completed. Reset and activation is required.
4 – 7	gRS1	Reserved (zeros)

Register: POSITION REQUEST ECHO (FINGER A IN INDIVIDUAL MODE)
Address: Byte 3

BIT	NAME	DESCRIPTION
0 – 7	gPRA	Echo of the requested position for the Gripper (or finger A in individual mode) 0x00 (Full Opening) to 0xFF (Full Closing)

Register: FINGER A POSITION
Address: Byte 4

BIT	NAME	DESCRIPTION
0 – 7	gPOA	Position of Finger A 0x00 (Fully opened) to 0xFF (Fully closed)

Register: FINGER A CURRENT
Address: Byte 5

BIT	NAME	DESCRIPTION
0 – 7	gCUA	Current of Finger A 0.1 * Current (in mA)

Register: FINGER B POSITION REQUEST ECHO
Address: Byte 6

BIT	NAME	DESCRIPTION
0 – 7	gPRB	Echo of the requested position for finger B 0x00 (Full Opening) to 0xFF (Full Closing)

Register: FINGER B POSITION
Address: Byte 7

BIT	NAME	DESCRIPTION
0 – 7	gPOB	Position of Finger B 0x00 (Fully opened) to 0xFF (Fully closed)

Register: FINGER B CURRENT
Address: Byte 8

BIT	NAME	DESCRIPTION
0 – 7	gCUB	Current of Finger B 0.1 * Current (in mA)

Register: FINGER C POSITION REQUEST ECHO
Address: Byte 9

BIT	NAME	DESCRIPTION
0 – 7	gPRC	Echo of the requested position for finger C 0x00 (Full Opening) to 0xFF (Full Closing)

Register: FINGER C POSITION
Address: Byte 10

BIT	NAME	DESCRIPTION
0 – 7	gPOC	Position of Finger C 0x00 (Fully opened) to 0xFF (Fully closed)

Register: FINGER C CURRENT
Address: Byte 11

BIT	NAME	DESCRIPTION
0 – 7	gCUC	Current of Finger C 0.1 * Current (in mA)

Register: SCISSOR POSITION REQUEST ECHO
Address: Byte 12

BIT	NAME	DESCRIPTION
0 – 7	gPRS	Echo of the requested position for the scissor axis 0x00 (Full Opening) to 0xFF (Full Closing)

Register: SCISSOR POSITION
Address: Byte 13

BIT	NAME	DESCRIPTION
0 – 7	gPOS	Position of the scissor axis 0x00 (Fully opened) to 0xFF (Fully closed)

Register: SCISSOR CURRENT
Address: Byte 14

BIT	NAME	DESCRIPTION
0 – 7	gCUS	Current for the scissor axis 0.1 * Current (in mA)

4.8 Example

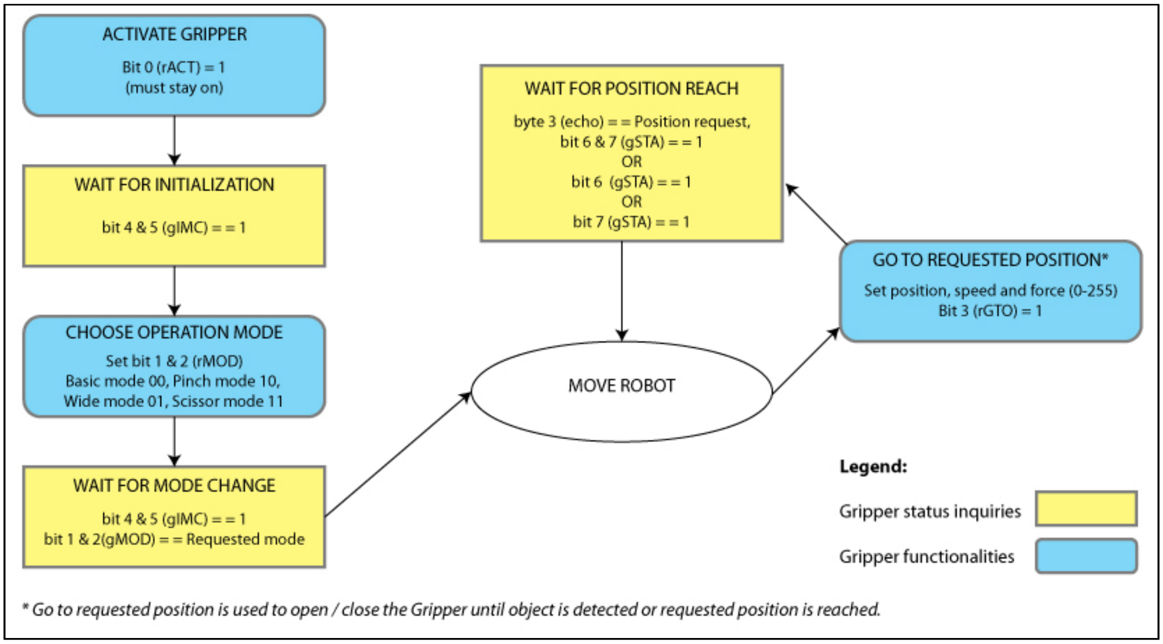

Figure 4.8.1 : Example of Adaptive Gripper S model registers.

4.9 MODBUS RTU communication protocol

The Robotiq Adaptive Gripper S model can be controlled over RS232 using the Modbus RTU protocol. This section is intended to provide guidelines for setting up a Modbus scanner that will adequately communicate with the gripper.

For a general introduction to Modbus RTU and for details regarding the CRC algorithm, the reader is invited to read the Modbus over serial line specification and implementation guide available http://www.modbus.org/docs/Modbus_over_serial_line_V1.pdf.

For debug purposes, the reader is also invited to download one of many free Modbus scanners such as the *CAS Modbus Scanner* from *Chipkin Automation Systems* available http://www.chipkin.com/cas-modbus-scanner.

4.9.1 Connection setup

The following table describes the connection requirement for controlling the Robotiq Adaptive Gripper S model using the Modbus RTU protocol.

PROPRIETY	VALUE
Physical Interface	RS232
Baud Rate	115,200 bps
Data Bits	8
Stop Bit	1
Parity	None
Number Notation	Hexadecimal
Supported Functions	Read Holding Registers (FC03) Preset Single Register (FC06) Preset Multiple Registers (FC16)
Exception Responses	Not supported
Slave ID	0x0009 (9)
Robot Output / Gripper Input First Register	0x03E8 (1000)
Robot Input / Gripper Output First Register	0x07D0 (2000)

Each register (word **-** 16 bits) of the Modbus RTU protocol is composed of **2** registers (bytes – 8 bits) from the Robotiq Adaptive Gripper S. The first Gripper output Modbus register (0x07D0) is composed from the first **2** Robotiq Adaptive Gripper S registers (byte 0 and byte 1).

4.9.2 Read holding registers (FC03)

Function code 03 (FC03) is used for reading the status of the Gripper (robot input). Examples of such data are Gripper status, object status, finger position, etc.

Ex: This message asks for register 0x07D0 (2000) and register 0x07D1 (2001) which contains Gripper Status, Object Detection, Fault Status and Position Request Echo.

Request is:

09 03 07 D0 00 02 C5 CE

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0002	Number of registers requested (2)
C5CE	Cyclic Redundancy Check (CRC)

Response is:

09 03 04 E0 00 00 00 44 33

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
E000	Content of register 07D0
0000	Content of register 07D1
4433	Cyclic Redundancy Check (CRC)

Note

The Adaptive Gripper register values are updated at a 200Hz frequency. It is therefore recommanded to send FC03 commands with a minimum delay of 5ms between them.

4.9.3 Preset single register (FC06)

Function code 06 (FC06) is used to activate functionalities of the Gripper (robot output). Examples of such data are action request, velocity, force, etc.

Ex: This message requests to initialize the Gripper by setting register 0x03E8 (1000), which contains Action Request and Gripper Options, to 0x0100.
Request is:

09 06 03 E8 01 00 09 62

where

BITS	DESCRIPTION
09	SlaveID
06	Function Code 06 (Preset Single Register)
03E8	Address of the register
0100	Value to write
0962	Cyclic Redundancy Check (CRC)

Response is an echo:

09 06 03 E8 01 00 09 62

where

BITS	DESCRIPTION
09	SlaveID
06	Function Code 06 (Preset Single Register)
03E8	Address of the register
0100	Value written
0962	Cyclic Redundancy Check (CRC)

4.9.4 Preset multiple registers (FC16)

Function code 06 (FC16) is used to activate functionalities of the Gripper (robot output). Examples of such data are action request, speed, force, etc.

Ex: This message requests to set position request, speed and force of the Gripper by setting register 0x03E9 (1001) and 0x03EA.

Request is:

09 10 03 E9 00 02 04 60 E6 3C C8 EC 7C

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E9	Address of the first register
0002	Number of registers to write
04	Number of data bytes to follow (2 registers x 2 bytes/register = 4 bytes)
00E6	Value to write to register 0x03E9
3CC8	Value to write to register 0x03EA
EC7C	Cyclic Redundancy Check (CRC)

Response is:

09 10 03 E9 00 02 91 30

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E9	Address of the first register
0002	Number of written
9130	Cyclic Redundancy Check (CRC)

4.9.5. Master read&write multiple registers (FC23)

Function code 23 (FC23) is used for reading the status of the Gripper (robot input) and activating functionalities of the Gripper (robot output) **simultaneously**. Examples of such data are Gripper status, object status, finger position, etc. Action requests are speed, force, etc.

Please refer to the C-Model instruction manual for a detailed example.

> ⚠ The C-Model example is only an illustration of the how-to for the S-Model, controls are not similar and bit addressing is not the same.

4.9.6 Modbus RTU example

This section depicts the example given in section 4.8 when programmed using the Modbus RTU protocol. The example is typical of a pick and place application. After activating the Gripper, the robot is moved to a pick-up location to grip an object. It moves again to a second location to release the gripped object.

Step 1: Activation Request

Request is:

09 10 03 E8 00 03 06 01 00 00 00 00 00 72 E1

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers to write to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0100	Value to write to register 0x03E9 (ACTION REQUEST = 0x01 and GRIPPER OPTIONS = 0x00): **rACT = 1 for "Activate Gripper"**
0000	Value to write to register 0x03EA
0000	Value to write to register 0x03EB
72E1	Cyclic Redundancy Check (CRC)

Response is:

09 10 03 E8 00 03 01 30

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)

Step 2: Read Gripper status until the activation is completed

Request is:

09 03 07 D0 00 01 85 CF

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0001	Number of registers requested (1)
85CF	Cyclic Redundancy Check (CRC)

Response (if the activation IS NOT completed):

09 03 02 11 00 55 D5

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
02	Number of data bytes to follow (1 registers x 2 bytes/register = 2 bytes)
1100	Content of register 07D0 (GRIPPER STATUS = 0x11, OBJECT STATUS = 0x00): gACT = 1 for "Gripper Activation", **gIMC = 1 for "Activation in progress"**
55D5	Cyclic Redundancy Check (CRC)

Response (if the activation IS completed):

09 03 02 31 00 4C 15

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
02	Number of data bytes to follow (1 registers x 2 bytes/register = 2 bytes)
3100	Content of register 07D0 (GRIPPER STATUS = 0x31, OBJECT STATUS = 0x00): gACT = 1 for "Gripper Activation", **gIMC = 3 for "Activation and mode change are completed"**
4C15	Cyclic Redundancy Check (CRC)

Step 3: Move the robot to the pick-up location

Step 4: Close the Gripper at full speed and full force

Request is:

09 10 03 E8 00 03 06 09 00 00 FF FF FF 42 29

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers to write to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0900	Value to write to register 0x03E9 (ACTION REQUEST = 0x09 and GRIPPER OPTIONS = 0x00): **rACT = 1 for "Activate Gripper"**, **rMOD=0 for "Go to Basic Mode"** , **rGTO = 1 for "Go to Requested Position"**
00FF	Value to write to register 0x03EA (GRIPPER OPTIONS 2 = 0x00 and POSITION REQUEST = 0xFF): **rPRA = 255/255 for full closing of the Gripper**
FFFF	Value to write to register 0x03EB (SPEED = 0xFF and FORCE = 0xFF): full speed and full force
4229	Cyclic Redundancy Check (CRC)

Response is:

09 10 03 E8 00 03 01 30

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)

Step 5: Read Gripper status until the grip is completed

Request is:

09 03 07 D0 00 08 45 C9

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0008	Number of registers requested (8)
45C9	Cyclic Redundancy Check (CRC)

Example of response if the grip **is not completed**:

09 03 10 39 C0 00 FF 08 0F 00 08 10 00 08 0F 00 89 00 00 73 70

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
10	Number of data bytes to follow (8 registers x 2 bytes/register = 16 bytes)
39C0	Content of register 0x07D0 (GRIPPER STATUS = 0x39, OBJECT STATUS = 0xC0): **gSTA = 0 for "Gripper is in motion towards requested position"**
00FF	Content of register 0x07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0xFF): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
080F	Content of register 0x07D2 (FINGER A POSITION = 0x08, FINGER A CURRENT = 0x0F): the position of finger A is 8/255 and the motor current is 150mA (these values will change during motion)

0008	Content of register 0x07D3 (FINGER B POSITION REQUEST ECHO = 0x00, FINGER B POSITION = 0x08)
1000	Content of register 0x07D4 (FINGER B CURRENT = 0x10, FINGER C POSITION REQUEST ECHO = 0x00)
080F	Content of register 0x07D5 (FINGER C POSITION = 0x08, FINGER C CURRENT = 0x0F)
0089	Content of register 0x07D6 (SCISSOR POSITION REQUEST ECHO = 0x00, SCISSOR POSITION = 0x89)
0000	Content of register 0x07D7 (SCISSOR CURRENT = 0x00)
7370	Cyclic Redundancy Check (CRC)

Example of response if the grip **is completed**:

09 03 10 B9 EA 00 FF BC 00 00 C1 00 00 BD 00 00 89 00 00 4E 17

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
10	Number of data bytes to follow (8 registers x 2 bytes/register = 16 bytes)
B9EA	Content of register 0x07D0 (GRIPPER STATUS = 0xB9, OBJECT STATUS = 0xEA): **gSTA = 2 for "Gripper is stopped. All fingers stopped before requested position"**, **gDTA = gDTB = gDTC = 2 for "Finger X has stopped due to a contact while closing"**
00FF	Content of register 0x07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0xFF): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
BC00	Content of register 0x07D2 (FINGER A POSITION = 0xBC, FINGER A CURRENT = 0x00): the position of finger A is 188/255 and the motor current is 0mA

00C1	Content of register 0x07D3 (FINGER B POSITION REQUEST ECHO = 0x00, FINGER B POSITION = 0xC1)
0000	Content of register 0x07D4 (FINGER B CURRENT = 0x00, FINGER C POSITION REQUEST ECHO = 0x00)
BD00	Content of register 0x07D5 (FINGER C POSITION = 0xBD, FINGER C CURRENT = 0x00)
0089	Content of register 0x07D6 (SCISSOR POSITION REQUEST ECHO = 0x00, SCISSOR POSITION = 0x89)
0000	Content of register 0x07D7 (SCISSOR CURRENT = 0x00)
4E17	Cyclic Redundancy Check (CRC)

Step 6: Move the robot to the release location

Step 7: Open the Gripper at full speed and full force

Request is:

09 10 03 E8 00 03 06 09 00 00 00 FF FF 72 19

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of registers to write to
06	Number of data bytes to follow (3 registers x 2 bytes/register = 6 bytes)
0900	Value to write to register 0x03E9 (ACTION REQUEST = 0x09 and GRIPPER OPTIONS = 0x00): **rACT = 1 for "Activate Gripper"**, **rMOD=0 for "Go to Basic Mode"** , **rGTO = 1 for "Go to Requested Position"**
0000	Value to write to register 0x03EA (GRIPPER OPTIONS 2 = 0x00 and POSITION REQUEST = 0x00): **rPR = 0/255 for full opening of the Gripper (partial opening would also be possible)**
FFFF	Value to write to register 0x03EB (SPEED = 0xFF and FORCE = 0xFF): full speed and full force
7219	Cyclic Redundancy Check (CRC)

Response is:

09 10 03 E8 00 03 01 30

where

BITS	DESCRIPTION
09	SlaveID
10	Function Code 16 (Preset Multiple Registers)
03E8	Address of the first register
0003	Number of written registers
0130	Cyclic Redundancy Check (CRC)

Step 8: Read gripper status until the opening is completed

Request is:

09 03 07 D0 00 08 45 C9

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
07D0	Address of the first requested register
0008	Number of registers requested (8)
45C9	Cyclic Redundancy Check (CRC)

Example of response if the opening **is not completed**:

09 03 10 39 C0 00 00 B8 0B 00 BD 0E 00 BA 0B 00 89 00 00 10 85

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)

10	Number of data bytes to follow (8 registers x 2 bytes/register = 16 bytes)
39C0	Content of register 0x07D0 (GRIPPER STATUS = 0x39, OBJECT STATUS = 0xC0): **gSTA = 0 for "Gripper is in motion towards requested position"**
0000	Content of register 0x07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0x00): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
B80B	Content of register 0x07D2 (FINGER A POSITION = 0xB8, FINGER A CURRENT = 0x0B): the position of finger A is 184/255 and the motor current is 170mA (these values will change during motion)
00BD	Content of register 0x07D3 (FINGER B POSITION REQUEST ECHO = 0x00, FINGER B POSITION = 0xBD)
0E00	Content of register 0x07D4 (FINGER B CURRENT = 0x0E, FINGER C POSITION REQUEST ECHO = 0x00)
BA0B	Content of register 0x07D5 (FINGER C POSITION = 0xBA, FINGER C CURRENT = 0x0B)
0089	Content of register 0x07D6 (SCISSOR POSITION REQUEST ECHO = 0x00, SCISSOR POSITION = 0x89)
0000	Content of register 0x07D7 (SCISSOR CURRENT = 0x00)
1085	Cyclic Redundancy Check (CRC)

Example of response if the opening **is completed**:

09 03 10 F9 FF 00 00 07 00 00 06 00 00 06 00 00 89 00 00 34 8D

where

BITS	DESCRIPTION
09	SlaveID
03	Function Code 03 (Read Holding Registers)
10	Number of data bytes to follow (8 registers x 2 bytes/register = 16 bytes)

F9FF	Content of register 0x07D0 (GRIPPER STATUS = 0xF9, OBJECT STATUS = 0xFF): **gSTA = 3 for "Gripper is stopped. All fingers reached requested position"**
0000	Content of register 0x07D1 (FAULT STATUS = 0x00, POSITION REQUEST ECHO = 0x00): the position request echo tells that the command was well received and that the GRIPPER STATUS is valid.
0700	Content of register 0x07D2 (FINGER A POSITION = 0x07, FINGER A CURRENT = 0x00): the position of finger A is 7/255 and the motor current is 0mA
0006	Content of register 0x07D3 (FINGER B POSITION REQUEST ECHO = 0x00, FINGER B POSITION = 0x06)
0000	Content of register 0x07D4 (FINGER B CURRENT = 0x00, FINGER C POSITION REQUEST ECHO = 0x00)
0600	Content of register 0x07D5 (FINGER C POSITION = 0x06, FINGER C CURRENT = 0x00)
0089	Content of register 0x07D6 (SCISSOR POSITION REQUEST ECHO = 0x00, SCISSOR POSITION = 0x89)
0000	Content of register 0x07D7 (SCISSOR CURRENT = 0x00)
348D	Cyclic Redundancy Check (CRC)

Step 9: Loop back to step 7 if other objects have to be gripped.